Final Project Summary Report

Interpretable Machine Learning for Predicting GHS Chemical Hazard Classifications: A Multi-Label Classification Approach Using PubChem Molecular Descriptors

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1-2. Dataset and feature matrix

Quantity	Value
Raw compound records harvested from PubChem	243,707
Removed: invalid SMILES	14
Removed: duplicate structures (by InChIKey)	370
Removed: no hazard label assigned	0
Final cleaned dataset	**243,323 compounds**
Compounds with a conflicting source vote	2,307
Multi-label compounds (more than one hazard)	63,781
Modelling subset (hardware-constrained)	40,000
Descriptors computed per compound	1,218
Removed by variance filter	402
Final feature matrix	**243,323 x 816**
Missing descriptor values imputed (median)	0
Split method	Bemis-Murcko scaffold split
Split sizes (train / val / test)	194,619 / 24,352 / 24,352
Distinct Bemis-Murcko scaffolds	57,591

3-4. Best algorithm and per-class performance

Best performing algorithm overall: XGBoost (mean AUC-ROC 0.9146 across the nine hazard classes on the scaffold-split test set). Mean AUC for every algorithm tested: XGBoost 0.9146, RandomForest_NoClassWeight 0.9078, RandomForest 0.8900, SVM 0.8711.

GHS column	Code	Authoritative meaning	AUC-ROC
GHS01_Explosive	GHS01	Explosive (exploding bomb)	0.9859
GHS02_Flammable	GHS02	Flammable (flame)	0.9708
GHS03_Oxidising	GHS03	Oxidiser (flame over circle)	0.9267
GHS04_CompressedGas	GHS04	Compressed gas (gas cylinder)	0.9951
GHS05_Corrosive	GHS05	Corrosive (corrosion)	0.8924
GHS06_AcuteToxicity	GHS06	Acute toxicity (skull and crossbones)	0.8296
GHS07_Irritant	GHS07	Irritant / harmful (exclamation mark)	0.8336
GHS08_HealthHazard	GHS08	Serious health hazard (health hazard)	0.9004
GHS09_Environmental	GHS09	Environmental hazard (environment)	0.8969

Note on column naming. Column names follow the official United Nations pictogram numbering. Three columns were renamed from the original study design, whose descriptive suffixes did not match that scheme: GHS07_HealthHazard → GHS07_Irritant; GHS08_Environmental → GHS08_HealthHazard; GHS09_Irritant → GHS09_Environmental. The underlying data were bound to the numeric pictogram code throughout and were therefore unaffected by the correction - no model was retrained and no classification changed.

5. Most predictive molecular features

Rank	Descriptor	Mean |SHAP| across all classes
1	LabuteASA	0.277935
2	TPSA	0.272887
3	Kappa2	0.263687
4	Chi4n	0.261146
5	BertzCT	0.240322
6	Chi3n	0.207268
7	MolLogP	0.169535
8	ExactMolWt	0.158715
9	MolWt	0.151152
10	Chi2n	0.132757

Top three overall: LabuteASA, TPSA, Kappa2

6. Malaysian validation performance

Sector	Compounds	Label accuracy	Hazard recall
Johor 2019 Emergency	12	0.907	0.836
Palm Oil Processing	7	0.905	0.900
Petrochemicals	9	0.876	0.839
Rubber Processing	7	0.762	0.708
Semiconductor (Penang)	8	0.736	0.658

9. Training time per algorithm

Algorithm	Training time
RandomForest	17.50 minutes
XGBoost	8.56 minutes
SVM	3.42 minutes
RandomForest_NoClassWeight_ablation	27.49 minutes
TOTAL	57.42 minutes

Measured on an Intel Core i7-6500U (two physical cores) with 7.9 GB of RAM, training on 194,619 compounds described by 816 features.

8. Steps where a fallback was used

Step	Fallback applied or limitation encountered
STEP10	FALLBACK APPLIED (proposal step 10a): SHAP values are computed on a random 500-compound sample of the 24,352-compound test set. Explaining every test compound would exceed this machine's 7.9 GB of RAM. This limitation is reported in the Methods section.
STEP10	FALLBACK APPLIED (proposal step 10a): SHAP values are computed on a random 500-compound sample of the 24,352-compound test set. Explaining every test compound would exceed this machine's 7.9 GB of RAM. This limitation is reported in the Methods section.
STEP2	batch starting at index 42600 failed; retrying compound-by-compound is skipped to save time
STEP2	batch starting at index 70800 failed; retrying compound-by-compound is skipped to save time
STEP2	batch starting at index 112500 failed; retrying compound-by-compound is skipped to save time
STEP2	batch starting at index 114600 failed; retrying compound-by-compound is skipped to save time
STEP2	batch starting at index 115200 failed; retrying compound-by-compound is skipped to save time
STEP2	batch starting at index 129900 failed; retrying compound-by-compound is skipped to save time
STEP2	batch starting at index 203700 failed; retrying compound-by-compound is skipped to save time
STEP2	time budget reached; CAS retrieved for 157,921 of 243,707 compounds. CAS is metadata only and is not used as a model feature, so this does not affect results.
STEP4	No issues. RDKit succeeded; mordred and PaDEL fallbacks were implemented but not needed.
STEP6	GHS01_Explosive: SMOTE would produce 388,942 rows, exceeding the 120,000 memory budget of this machine. Class weighting (6b) is used for this class instead.
STEP6	GHS02_Flammable: SMOTE would produce 372,226 rows, exceeding the 120,000 memory budget of this machine. Class weighting (6b) is used for this class instead.
STEP6	GHS03_Oxidising: SMOTE would produce 388,566 rows, exceeding the 120,000 memory budget of this machine. Class weighting (6b) is used for this class instead.
STEP6	GHS04_CompressedGas: SMOTE would produce 388,780 rows, exceeding the 120,000 memory budget of this machine. Class weighting (6b) is used for this class instead.
STEP6	GHS05_Corrosive: SMOTE would produce 319,898 rows, exceeding the 120,000 memory budget of this machine. Class weighting (6b) is used for this class instead.
STEP6	GHS06_AcuteToxicity: SMOTE would produce 364,916 rows, exceeding the 120,000 memory budget of this machine. Class weighting (6b) is used for this class instead.
STEP6	GHS07_Irritant: SMOTE would produce 352,330 rows, exceeding the 120,000 memory budget of this machine. Class weighting (6b) is used for this class instead.
STEP6	GHS08_HealthHazard: SMOTE would produce 367,466 rows, exceeding the 120,000 memory budget of this machine. Class weighting (6b) is used for this class instead.
STEP6	GHS09_Environmental: SMOTE would produce 360,860 rows, exceeding the 120,000 memory budget of this machine. Class weighting (6b) is used for this class instead.
STEP7	FALLBACK APPLIED (proposal step 7c): the SVM is trained on the top 100 features by Random Forest importance instead of all 817, because full-feature RBF training would exceed 30 minutes on this 2-core machine.
STEP7	ADDITIONAL HARDWARE FALLBACK: an RBF kernel matrix for 32,000 compounds would need about 8.2 GB, more than this machine has. The SVM is trained on a stratified subsample of 8,000 compounds. This is a hardware limitation and is reported in the Methods section; the SVM's scores are therefore not strictly comparable with those of RF and XGBoost.
STEP7	FALLBACK APPLIED (proposal step 7c): the SVM is trained on the top 100 features by Random Forest importance instead of all 816, because full-feature RBF training would exceed 30 minutes on this 2-core machine.
STEP7	ADDITIONAL HARDWARE FALLBACK: an RBF kernel matrix for 194,619 compounds would need about 303.0 GB, more than this machine has. The SVM is trained on a stratified subsample of 8,000 compounds. This is a hardware limitation and is reported in the Methods section; the SVM's scores are therefore not strictly comparable with those of RF and XGBoost.
STEP7	Step 6 applied no oversampling to any class - the projected matrices exceeded the memory budget at this dataset size. This ablation therefore measures the effect of removing class weighting, NOT the effect of SMOTE. The SMOTE comparison is only available at the 40,000-compound scale, where it was measured.
STEP8	Random Forest: one fit took 108.6s, so the proposal's n_iter=30 would need 3.6 hours. Reduced to n_iter=5 with cv=3 to fit the 10-minute budget.

Step	Fallback applied or limitation encountered
STEP8	XGBoost: one fit took 23.0s, so the proposal's n_iter=30 would need 0.8 hours. Reduced to n_iter=5 with cv=3 to fit the 1-minute budget.

Principal fallback, Step 1. No Python installation existed on the target machine, and the official python.org Windows installer failed twice with WiX bootstrapper exit code 0x3 because no administrator elevation was available for its chained MSI packages. conda was also absent, so the conda fallback named in the proposal could not be used. A standalone CPython 3.11.15 build was installed instead via the uv package manager, which requires no Windows installer. All subsequent package installation proceeded normally.

Two defects found and corrected during development. Both produced results that looked entirely plausible, which is why they are recorded here rather than quietly fixed. First, the scaffold group allocation starved the rare classes: only 30 per cent of compressed gases and 39 per cent of oxidisers reached the training partition instead of 80 per cent, because every acyclic molecule forms a single-compound scaffold group and those groups were allocated last. Allocation is now performed by group-wise iterative stratification, and every class lands within one percentage point of its intended share. Second, the modelling set had been restricted to 40,000 compounds on memory grounds; a controlled experiment showed this cost 0.055 of mean AUC-ROC, four times the bootstrap confidence interval, so all results now use the complete dataset.

7. Files produced

546 files totalling 9.4 GB. The twenty-five largest are listed below; the complete inventory is saved as FINAL_file_inventory.csv.

File	Size
colab_results\colab_X_full.npy	1.1 GB
features\STEP4_X.npy	757.4 MB
STEP6_X_train_balanced.npy	605.8 MB
features\STEP6_balanced\X_GHS01_Explosive.npy	605.8 MB
features\STEP6_balanced\X_GHS07_Irritant.npy	605.8 MB
features\STEP6_balanced\X_GHS06_AcuteToxicity.npy	605.8 MB
features\STEP6_balanced\X_GHS04_CompressedGas.npy	605.8 MB
features\STEP6_balanced\X_GHS05_Corrosive.npy	605.8 MB
features\STEP6_balanced\X_GHS08_HealthHazard.npy	605.8 MB
features\STEP6_balanced\X_GHS09_Environmental.npy	605.8 MB
features\STEP6_X_train_balanced.npy	605.8 MB
features\STEP6_balanced\X_GHS03_Oxidising.npy	605.8 MB
features\STEP6_balanced\X_GHS02_Flammable.npy	605.8 MB
models\STEP7_rf_model.pkl	103.0 MB
models\STEP7_rf_noclassweight_ablation.pkl	97.6 MB
models\STEP8_rf_tuned.pkl	90.4 MB
STEP3_cleaned_ghs_dataset.csv	76.5 MB
data\cleaned\STEP3_cleaned_ghs_dataset_20260805.csv	76.5 MB
data\cleaned\STEP3_cleaned_ghs_dataset.csv	76.5 MB
features\STEP4_feature_matrix.csv	65.6 MB
STEP4_feature_matrix.csv	65.6 MB
data\raw\STEP2_raw_ghs_dataset_20260805.csv	58.3 MB
STEP2_raw_ghs_dataset.csv	58.3 MB
data\raw\STEP2_raw_ghs_dataset.csv	58.3 MB
features\STEP4_compound_index.csv	43.3 MB

10. Recommended next steps

1. Train the support vector machine on the full feature set and full training partition. Its scores here are not strictly comparable with Random Forest and XGBoost because it was restricted to 100 features and a subsampled training set for tractability. Note that this is not simply a matter of buying more memory: an RBF kernel matrix for the full training partition would be roughly 300 GB, so a different formulation - a linear kernel, or Nystroem approximation - would be needed rather than a larger machine.

2. Investigate the threshold calibration for the irritant class. At the high prevalence this class reaches in the full dataset, the F1-optimal threshold degenerates towards predicting the positive class for almost every compound, giving a high F1 alongside a very low Matthews correlation coefficient. A prevalence-aware criterion, or simply reporting the safety-first threshold for this class, would be more honest.

3. Add a graph neural network baseline. Message-passing networks learn their own representation from the molecular graph and typically outperform fingerprint-based models on large chemical datasets; the scaffold split and evaluation code here would carry over unchanged.

4. Define and report an applicability domain. Predictions for compounds structurally remote from the training set should be flagged as extrapolation rather than reported with the same confidence. A distance-to-training-set measure in descriptor space would be the simplest implementation.

5. Incorporate Malaysian CLASS Regulations classifications into the training data. The current model learns from European, Japanese, Australian and American sources; adding Malaysian classifications would improve coverage of locally significant chemicals.

6. Extend coverage to inorganic and organometallic compounds. The descriptors used here are designed for covalent organic structures, and predictions for salts, metal oxides and coordination compounds are correspondingly less reliable.

7. Conduct a prospective validation. Hold back chemicals classified after a chosen date and test on those, which is the closest computational analogue of the real deployment scenario.

DISCLAIMER. The models described in this report are computational screening tools. They do not replace laboratory testing or regulatory assessment under Malaysia's Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013.